

EXAM 7 – SPRING 2015

24. (2.25 points)

For each of the risk measures identified below, provide a definition and explain one significant limitation of the measure in the context of an ERM application.

a. (0.75 point)

Value at Risk (VaR)

b. (0.75 point)

Tail Value at Risk (TVaR)

c. (0.75 point)

Risk-adjusted TVaR (RTVaR)

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

QUESTION 24	
TOTAL POINT VALUE: 2.25	LEARNING OBJECTIVE: C4: Demonstrate the properties of various risk measures and their limitations. C5: Describe how risk measures and risk modeling, including allocation, can affect strategic management.
SAMPLE ANSWERS	
Part a: 0.75 point	
<i>Sample Answers for the Definition</i> VaR at α is the specified value at the α percentile. $\text{VaR} = E[X \mid X = \alpha]$ <i>Sample Answers for the Limitation</i> It is a single point, so it does not provide much information on the distribution. It is not sub-additive, so does not provide diversification benefit. It is only one point, and does not account for risk in the tail, or below the VaR. These risks are important in a risk management context.	
Part b: 0.75 point	
<i>Sample Answers for the Definition</i> TVaR is the expected value of losses above a specified percentile of the distribution. $\text{TVaR} = E[X \mid X > \alpha]$ <i>Sample Answers for the Limitation</i> TVaR is linear in the tail and as a result does not reflect the risk averse attitude that a risk twice as large is considered more than twice as bad. This ignores risk below α, where the risks may not be extreme, but could still be significant.	
Part c: 0.75 point	
<i>Sample Answers for the Definition</i> $\text{RTVaR} = \text{TVaR} + c * \text{std dev}(X \mid X > \alpha)$, where c is some constant RTVaR is the TVaR plus some proportion of the standard deviation in the tail. <i>Sample Answers for the Limitation</i> A limitation of RTVaR is that it only provides a calculation based on the tail losses & does not evaluate all losses in the distribution, meaning it is not a complete risk measure. For a skewed distribution, the standard deviation loading may not provide enough weight in the tail. A criticism of RTVaR is that even with the inclusion of standard deviation it may not place sufficient weight on very large losses due to being a quadratic measure.	
EXAMINERS' REPORT	
Candidates scored generally well on this question. To receive full credit on each part, candidates had to (1) provide a valid definition for the given risk measure, either in words or in a formula and (2) provide a valid limitation to that risk measure, with a brief explanation. Candidates generally performed well on this question, though part c proved more challenging than parts a and b.	

EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

Part a
Candidates scored well on this part. The most common error was to state a limitation without any explanation (this earned partial credit).
Part b
Candidates also scored well on this part. Common errors included: • Stating as a limitation that the risk measure includes losses excess of insolvency. The paper mentions this, but goes on to explain that it is not actually a limitation. • Stating a limitation without explanation (this earned partial credit).
Part c
Candidates found this part more challenging than parts a and b. Common errors included: • Confusion with WTVAR. • Forgetting the constant “c” applied to the standard deviation in the formula. • Stating a limitation without explanation (this earned partial credit).